How-To Library

Informatica

Address Verification Best Practices for Czech Republic

© Copyright Informatica LLC 2020, 2021. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

This document tells you how to connect the elements in a Czech Republic address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in the Czech Republic.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

Table of Contents

Introduction. . . . 2
Process Modes. . . . 2
Character Set. . . . 3
Address Format and Address Structure. . . . 3
Key Address Elements. . . . 3
Mapping Input Address Data to Address Elements. . . . 3
Input and Result XML in Address Verification (On-Premises). . . . 4
Input and Result XML in Address Verification. . . . 5
Address Enrichments. . . . 6
Frequently Asked Questions. . . . 7
Reference Address Databases. . . . 7
Additional Documentation Resources. . . . 7

Introduction

Informatica Address Verification can validate and enhance addresses in the Czech Republic. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Czech Republic address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for Czech Republic addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE

- PARSE

Character Set

Czech Republic reference databases store address data in the Latin-2 script. By default, Informatica Address Verification returns the verified Czech Republic addresses in the Latin-2 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Czech Republic postal service recognizes the following structure for a postal address:

```
Line 1: STREET BUILDING NUMBER [/] HOUSE NUMBER
Line 2: POSTAL CODE CITY [SORTING CODE]
Line 3: COUNTRY
```

The following sample address illustrates the structure:

```
SEDLÁKOVA 44/12
301 00 PLZEN 1
CZE
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Postal Code
- Street

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name	SEDLÁKOVA	Street
Building Number	44	Building
House Number	12	Number or HouseNumber
Postal Code	301 00	PostalCode
City	PLZEN 1	Locality

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">CZECH REPUBLIC</Country>
        <Locality Item="1" Type="COMPLETE">PLZEN 1</Locality>
        <PostalCode Item="1" Type="UNFORMATTED">301 00</PostalCode>
        <Street Item="1" Type="COMPLETE">SEDLÁKOVA</Street>
        <Number Item="1" Type="COMPLETE">12</Number>
        <Building Item="1" Type="COMPLETE">44</Building>
    </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.16.0.55045" -->

<Result     ProcessStatus="C4"
     ModeUsed="BATCH"
     Count="1"
     CountOverflow="NO"
     CountryISO3="CZE"
     PreferredScript="LATIN"
     PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
     MailabilityScore="4"
     ResultPercentage="96.11"
     ElementResultStatus="F07880F0F000F00000E0"
     ElementInputStatus="60400060600060000060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="10100010100010000010"
     AddressType="B"
     LanguageISO3="???">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">CZECH REPUBLIC</Country>
        <Locality Item="1">Plzen 3</Locality>
        <Locality Item="2">Valcha</Locality>
        <PostalCode Item="1">301 00</PostalCode>
        <Province Item="1">Plzensky</Province>
        <Street Item="1">Sedlakova</Street>
        <Number Item="1">12</Number>
        <Building Item="1">44</Building>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Sedlakova 44/12</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">301 00 Plzen 3</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Sedlakova 44/12</FormattedAddressLine>
        <FormattedAddressLine Line="2">Valcha</FormattedAddressLine>
        <FormattedAddressLine Line="3">301 00 Plzen 3</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>Sedlakova 44/12
Valcha
301 00 Plzen 3</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Czech Republic address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
    <soap:Header/>
    <soap:Body>
        <v:Process>
          <v:login>****</v:login>
          <v:password>****</v:password>
          <v:parameters>
            <v:ProcessMode>BATCH</v:ProcessMode>
          <v:ServiceParameters></v:ServiceParameters>
          <v:ValidationParameters>
          </v:ValidationParameters>
          </v:parameters>
          <v:addresses>
            <v:Address>
             <v:FormattedAddress>
                <v:string>SEDLÁKOVA 44/12</v:string>
                <v:string>301 00 PLZEN 1</v:string>
                <v:string>CZECH REPUBLIC</v:string>
             </v:FormattedAddress>
            </v:Address>
          </v:addresses>
          <v:enrichments>
          </v:enrichments>
        </v:Process>
    </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>C4</ProcessStatus>
                  <CountryISO3>CZE</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>4</MailabilityScore>
                        <ResultPercentage>96.11</ResultPercentage>
                        <ElementInputStatus>6040006060006000060</ElementInputStatus>
                        <ElementResultStatus>F07880F0F000F00000E0</ElementResultStatus>
                        <ElementRelevance>1010001010001000010</ElementRelevance>
                        <ExtElementStatus>00000000000000000000</ExtElementStatus>
                        <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                        <AddressType>B</AddressType>
                        <LanguageISO3>???</LanguageISO3>
                        <Address>
                           <Building>
                              <string>44</string>
                           </Building>
                           <Street>
                              <string>Sedlákova</string>
                           </Street>
                           <HouseNumber>
                              <string>12</string>
                           </HouseNumber>
                           <Locality>
                              <string>Plzeň 3</string>
```

```
                              <string>Valcha</string>
                           </Locality>
                           <PostalCode>
                              <string>301 00</string>
                           </PostalCode>
                           <Province>
                              <string>Plzeňský</string>
                           </Province>
                           <Country>
                              <string>CZECH REPUBLIC</string>
                           </Country>
                           <DeliveryAddressLines>
                              <string>Sedlákova 44/12</string>
                           </DeliveryAddressLines>
                           <CountrySpecificLocalityLine>
                              <string>301 00 Plzeň 3</string>
                           </CountrySpecificLocalityLine>
                           <FormattedAddress>
                              <string>Sedlákova 44/12</string>
                              <string>Valcha</string>
                              <string>301 00 Plzeň 3</string>
                              <string>CZECH REPUBLIC</string>
                           </FormattedAddress>
                           <AddressComplete>Sedlákova 44/12;Valcha;301 00 Plzeň 3;CZECH
REPUBLIC</AddressComplete>
                        </Address>
                     </ResultData>
                  </ResultDataSet>
               </Result>
            </Results>
         </ProcessResult>
      </ProcessResponse>
   </soap:Body>
</soap:Envelope>
```


Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Czech Republic addresses.

The following table lists the types of enrichment that you can add to a Czech Republic address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Czech Republic addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Country-Specific Address Enrichments	You can retrieve the RUIAN ID codes as enrichments for valid Czech Republic addresses. The RUIAN ID enrichment comprises the following codes: - RUIANAM_ID. Uniquely identifies the address delivery point. - RUIANSO_ID. Identifies the address to building level. - RUIANTEA_ID. Identifies the building entrance.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Czech Republic addresses.
Arrival Point Geocoding	You can retrieve arrival point geocoordinates for valid Czech Republic addresses.

Frequently Asked Questions

Why does Address Verification return a digit in the Locality field with some city names?

Address Verification returns a digit after a city name in the Locality field if the city uses sorting codes to define districts. In this case, the digit belongs in the Locality field.

Why does Address Verification return Locality 2 information in the complete address?

Address Verification returns Locality 2 information in the complete address because the reference data contains the information. The information is not required in the postal address.

How does Informatica Address Verification handle building number and house number combinations?

A Czech Republic address can contain a single house number, or it may contain a building number and house number in combination. The presence or absence of the building number depends on the history of the street.

For example, in the following address, 77 is the building number and 3 is the house number:

```
17. LISTOPADU 77/3
110 00 PRAHA 1
```

Note: 17. LISTOPADU is the street name (17th November).

Both the building number and the house number appear once on a given street. A given building number does not occur with multiple house numbers, and likewise a house number does not occur with multiple building numbers. If you submit an address in batch or interactive mode that uses both a building number and a house number, but you omit either number, Address Verification returns the complete address and a C4 process status. In interactive mode, Address Verification returns a single address, as the input address finds a unique match in the reference data. An address that contains both a building number and house number can exist in the same street as an address that contains a house number only.

Reference Address Databases

Informatica updates the Czech Republic reference address databases twice a year. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the Informatica DaaS Newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Czech Republic addresses:

- `CZE5BI.MD`. Batch and interactive data.
- `CZE5CA.MD`. CAMEO data.
- `CZE5E1.MD`. Supplementary data.
- `CZE5FC.MD`. Fast completion data.
- `CZE5GC.MD`. Standard geocoding data.
- `CZE5GCAP.MD`. Arrival point geocoding data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification uses a cloud interface.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

Author

Shahani Natalia Mendonca

Acknowledgements

The author would like to thank Amarpal Kaur Sohi, David Handy, Andre Kirsten, Farhan Ashraf, Matthew Hall, and Xiayi Zhang for their assistance.